

Do Gaussian Splats Keep the Nuclei?

An Object-Level Audit of Gaussian Volume Compression for Fluorescence Microscopy

[TODO: author names]

[TODO: affiliation]

Abstract—Gaussian splats are being adopted to store and stream large 3D microscopy and medical volumes, with budgets chosen and quality reported by PSNR. We ask whether, at a given number of bytes on disk, standard analysis still finds the nuclei. On four densely annotated frames from two *C. elegans* embryos (1095 labelled nuclei), we compare a Gaussian-splat compressor (Luxar) with JPEG-XL, JPEG2000, ZFP and downsampling at matched stored bytes, scoring a blob detector and a learned segmenter (Cellpose) against human labels, under decision rules fixed before the replication. Luxar’s self-calibrated budget ($\approx 28\text{--}30\times$) keeps $\geq 97\%$ of nuclei under both detectors, but at a ratio where JPEG-XL is also nearly lossless ($\geq 92\%$ up to $82\text{--}118\times$) and far cheaper to compute. At hand-chosen budgets of $\approx 200\times$ and beyond, splats keep far more nuclei than JPEG2000 (Cellpose: $82\text{--}88\%$ vs. $32\text{--}74\%$), until a cliff at $\lesssim 2.3$ splats per nucleus that PSNR, flat within 19.1–22.6 dB, does not reveal. Splat reconstructions also cost the learned segmenter 6–15 points of recall relative to JPEG-XL at $45\times$ to $94\times$ on every frame, while the blob detector is unaffected. Object-level evaluation with the downstream detector should accompany Gaussian volume compression.

I. Introduction

Light-sheet and confocal time-lapse microscopy produce terabytes per experiment, and medical archives face the same pressure. Representing a volume as a set of anisotropic 3D Gaussians [1] is attractive for such data: it is compact, continuous and renders directly on the GPU. Recent systems fit Gaussian splats to microscopy volumes at scale [2], to slice-based microscopy and ultrasound [3], to medical volumes for compact storage [4], [5], and to tomographic reconstruction [6], [7]. Quality is measured by PSNR or SSIM throughout, and budgets are chosen by PSNR: Luxar selects its splat count by self-supervised held-out PSNR [8] and describes its fits as intended for visualisation rather than quantitative analysis [2].

What a microscopist needs from a compressed volume is that the analysis still works. Whether a nucleus remains detectable hinges on a small, often faint peak. Radiology learned this for lossy codecs, where acceptability is judged by whether lesions remain detectable, not by PSNR [9]. Gaussian volume compression has not been held to that standard.

Contributions. (i) A rate-detectability protocol: matched bytes on disk, dense human labels, two detectors, one-to-one matching in physical units, and decision rules committed before a replication on a

second embryo. (ii) Luxar’s own budget is safe but sits where JPEG-XL is equally safe; its advantage over codecs appears only at hand-chosen budgets $\gtrsim 200\times$, bounded by a cliff that PSNR does not show. (iii) Splat compression lowers a learned segmenter’s recall at ratios where JPEG-XL does not—a detector dependence the codecs do not have.

II. Related Work

Gaussian mixtures have long modelled nuclei for segmentation and lineage tracking [10]; our question concerns Gaussians as a storage format. Gaussian splatting [1] has been adapted to CT [7], [6], to slice-to-volume MRI with closed-form point-spread functions [11], and to volume compression [2], [3], [4], [5], each evaluated by image-fidelity metrics. JPEG2000 [12], JPEG-XL [13] and the error-bounded ZFP [14] are the natural baselines at matched size. We use Cell Tracking Challenge labels [15], [16] and two standard detectors: a scale-normalised Laplacian of Gaussian [17] and Cellpose [18].

III. Methods

Data. Fluo-N3DH-CE [16]: *C. elegans* embryos imaged at $1.0 \times 0.09 \times 0.09 \mu\text{m}$ (8-bit, ≈ 12 MB per frame), with every nucleus labelled. We use embryo 1 (sequence 01) frames t150 and t194 and embryo 2 (sequence 02) frames t150 and t180, with 192 and 362 and 190 and 351 labelled nuclei; embryo 1 t100 (93 nuclei) enters only the budget analysis. Nuclei measure $3.6 \mu\text{m}$ (median over the challenge’s segmentation masks); all spatial parameters are fixed multiples of this diameter D .

Compressors. Luxar [2] with default settings, 1000 iterations and anisotropic voxel size, over seed budgets K from 10^3 to 6.4×10^4 (embryo 2: 3.2×10^4); stored size is its .gsplats.zarr on disk. For embryo 1 we also run Luxar’s own calibration—blind-spot held-out PSNR [8]—over the same K grid to obtain its recommended K^* . JPEG2000 (3D, OpenJPEG), JPEG-XL (per slice), ZFP fixed-rate and trilinear downsampling are bisected to each target size; sizes a codec cannot reach are flagged and excluded from size-matched comparisons.

Detectors and scoring. (a) LoG: local maxima in a physically isotropic window, ranked by scale-normalised LoG response ($\sigma \in \{0.2, 0.27, 0.33, 0.4\}D$); the top N are

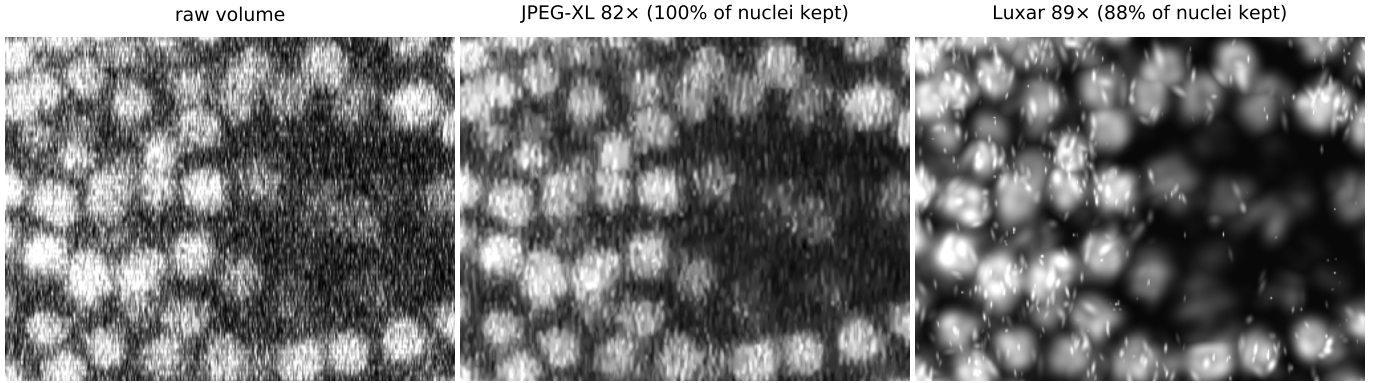


Fig. 1. The same region of embryo 1, t194 (3-slice maximum projection) from the raw volume, JPEG-XL and Luxar at similar size; titles give whole-frame nuclei kept by Cellpose. JPEG-XL preserves the textured appearance of nuclei; Luxar renders smooth blobs on a black background with thin streaks from elongated splats—an image distribution a learned segmenter was not trained on.

TABLE I
Nuclei kept near matched ratios, mean over the four frames (superscript: fewer frames with a size match). Match radius $0.6D$.

Ratio	LoG			Cellpose		
	JXL	J2K	Luxar	JXL	J2K	Luxar
$\approx 30\times$	1.00	1.02	1.01 ²	0.99	0.98	0.97 ²
$\approx 50\times$	1.00	1.00	1.02	0.98	0.91	0.88
$\approx 90\times$	1.00	0.98 ²	1.01	0.97	0.99 ²	0.88
$\approx 130\times$	1.01 ²	0.95	1.01	0.95 ²	0.91	0.85
$\approx 230\times$	—	0.64	1.00	—	0.55	0.84

kept, N = number of labelled nuclei. (b) Cellpose 3 nuclei, 2D per slice stitched across z , at its own operating point. Detections match labels one-to-one within $0.6D$ (Table II varies it). Nuclei kept = nuclei found after compression / found in the raw volume by the same detector (1.0 = nothing lost). Survival, a label-free measure, is the fraction of the raw volume’s 100 strongest LoG detections still found.

Decision rules. Before the embryo-2 replication and the calibration runs, we committed written pass/fail rules for each claim [TODO: repository link]. One claim failed as worded and is reported in its narrowed form below.

IV. Results

Codecs. JPEG-XL is nearly lossless for detection up to its maximum ratio (82–118 $\times$; worst case 92% under Cellpose). JPEG2000 degrades beyond $\approx 100\times$ and keeps 32–83% of nuclei at $\approx 230\times$ (Fig. 2, Table I); downsampling and fixed-rate ZFP fail early.

Luxar at its own budget. Luxar’s calibration recommends $K^* = 64,000$ on both embryo-1 frames—the top of our grid, with held-out PSNR still rising. There it keeps 101% (LoG) and 97–98% (Cellpose) of nuclei, at 28–30 $\times$. Its own rule is therefore safe here, but it lands at ratios where JPEG-XL is equally safe, while fitting takes 17–46 min per volume on an A100 (12–27 min on an H100) against seconds for the codecs.

Hand-chosen budgets. Below K^* , splats reach ratios no codec matches. At $\approx 230\times$ Luxar keeps 96–103% (LoG) and 82–88% (Cellpose) of nuclei, against 48–83% and 32–74% for JPEG2000, on both embryos; 95% bootstrap intervals over nuclei exclude zero on 4 of 4 frames for LoG but only 3 of 4 for Cellpose (embryo 1 t150 is unresolved). The advantage is not uniform: at 110 $\times$ to 135 $\times$ on embryo 2, JPEG2000 at 110 $\times$ keeps 96–102% under Cellpose versus 87% for Luxar, so our pre-registered claim “Luxar leads beyond 120 $\times$ ” is narrowed to $\approx 200\times$ and beyond. Loss then sets in abruptly once splats become scarce (Fig. 3): on all four frames Luxar drops below 90% (LoG) only at ≤ 2.3 splats per labelled nucleus, reaching 66–67% at one per nucleus. Over the same budgets Luxar’s PSNR moves only within 19.1–22.6 dB (Fig. 4); across all reconstructions PSNR orders 7–24% of pairs more than 1 dB apart the wrong way round for faint nuclei kept.

Detector dependence. With the blob detector, Luxar keeps 98–104% of nuclei at 45 $\times$ to 94 $\times$; with Cellpose only 83–92%, 6–15 points below JPEG-XL at the same ratios, on all 4 frames (95% bootstrap intervals exclude zero on 4 of 4, one only narrowly). The two detectors need very different budgets (Fig. 3): the blob detector recovers every nucleus once there are ≈ 3 –5 splats per nucleus, whereas Cellpose stays at 80–90% from ≈ 5 to 50 splats per nucleus and approaches the codecs only at ≈ 100 or more—the regime of Luxar’s own K^* . Covering each nucleus is not enough for a learned segmenter; the reconstruction must also look like the images it was trained on (Fig. 1). The deficit is moderate relative to Cellpose’s own variability—its scores move by up to 14 points between neighbouring sizes of the same codec—but consistent in sign across frames and embryos, and the codecs show no comparable detector dependence.

A label-free guard. Survival, which needs no labels, tracks nuclei kept across Luxar fits (Spearman $\rho = 0.68$, $n = 30$). Being itself a blob-detector measure, it can flag the budget cliff but not the segmenter’s appearance gap.

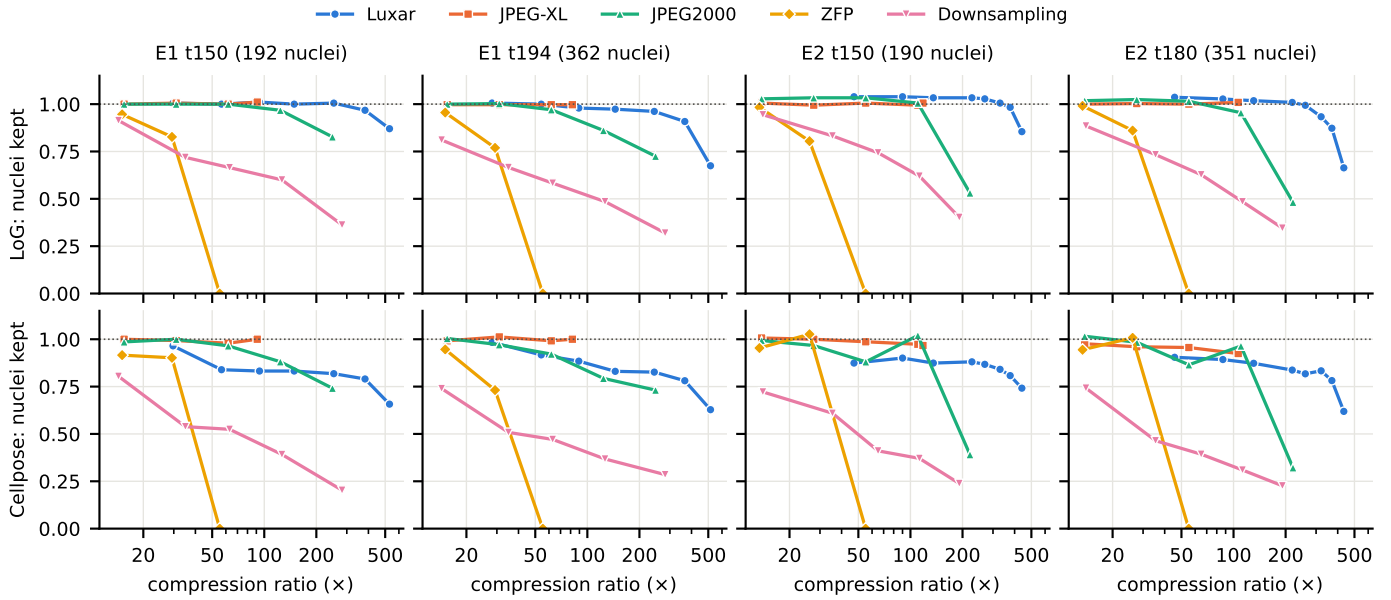


Fig. 2. Nuclei kept (relative to the raw volume) against compression ratio at matched bytes on disk, for the LoG blob detector (top) and Cellpose (bottom), on four frames from two embryos (E1, E2). JPEG-XL cannot exceed 82–118 $\times$ on this data.

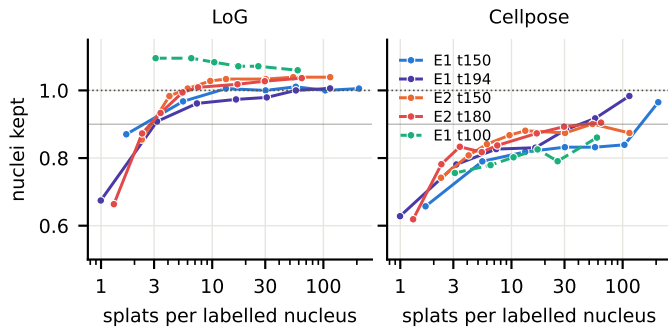


Fig. 3. The budget cliff: nuclei kept against splats per labelled nucleus for every Luxar fit (dashed: E1 t100, whose smallest budget has 3.1 splats per nucleus).

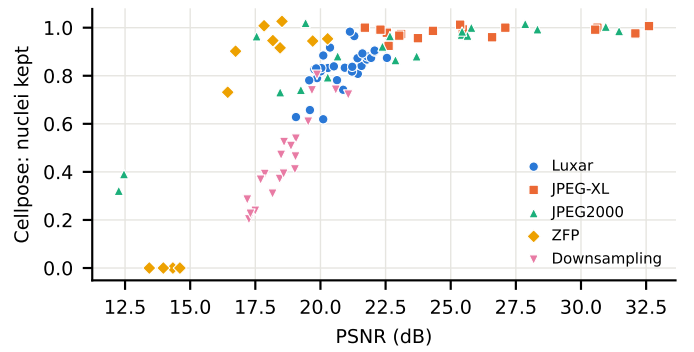


Fig. 4. PSNR against nuclei kept (Cellpose) for every reconstruction on the four frames.

TABLE II

Match-radius sensitivity: nuclei found in the raw volumes (sum over four frames) and nuclei kept near 90 $\times$ (mean).

Radius	LoG			Cellpose		
	raw	JXL	Luxar	raw	JXL	Luxar
0.4 D	1009	1.00	1.02	658	1.01	0.84
0.6 D	1032	1.00	1.01	788	0.97	0.88
0.8 D	1034	1.00	1.01	796	0.99	0.90

V. Discussion and Limitations

On this data, the case for Gaussian-splat storage rests on ratios well beyond JPEG-XL’s reach, and there it depends on two things that image-fidelity metrics hide: the budget, and the detector used afterwards. We recommend (i) JPEG-XL wherever its ratio suffices; (ii) for higher ratios, checking a splat budget at the object level before

trusting it (survival is a label-free start); and (iii) reporting rate-detectability with the detector that will be used downstream.

Limitations: one organism (two embryos, five frames); Cellpose in its 2D-stitched mode and with the variability noted above; a single Gaussian compressor at default settings; the calibration evaluated on embryo 1 only; and larger early-embryo nuclei (t100) that fit our fixed D less well. Fine-tuning segmenters on splat reconstructions, and the same audit for small low-contrast lesions in CT, are future work.

VI. Conclusion

Measured at matched bytes against human labels, Gaussian-splat compression keeps nuclei at ratios no standard codec reaches, but only above a budget that PSNR cannot identify, and with a recall cost for learned

segmenters that no image-fidelity metric exposes. Object-level, detector-specific rate–detectability evaluation should accompany Gaussian volume compression.

References

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3D gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] L. A. Royer, “Luxar: Gaussian splatting for microscopy and scalable interactive web visualisation of multidimensional scientific data,” Zenodo, <https://zenodo.org/records/22912050>, 2026, preprint; software at <https://github.com/royerlab/luxar>.
- [3] D. Kong, Y. Wang, W. Guo, Y. Bu, B. Zhang, Y. Duan, X. Yue, W. Du, Y. Zhong, Y. Chen, and C. Ma, “GaussianPile: A unified sparse Gaussian splatting framework for slice-based volumetric reconstruction,” *arXiv:2603.20611*, 2026.
- [4] X. Li and Q. Gu, “GSToken: Geometry-structured Gaussian tokens for compact 3D medical image representation,” *arXiv:2608.17425*, 2026.
- [5] M. R. Martin, J. Insley, V. A. Mateevitsi, S. Rizzi, and K.-L. Ma, “Fixed-budget Gaussian volume encoding with structure-aware allocation,” *arXiv:2608.14112*, 2026.
- [6] Y. Li, X. Fu, H. Li, S. Zhao, R. Jin, and S. K. Zhou, “3DGR-CT: Sparse-view CT reconstruction with a 3D Gaussian representation,” *Medical Image Analysis*, vol. 103, p. 103585, 2025.
- [7] R. Zha, T. J. Lin, Y. Cai, J. Cao, Y. Zhang, and H. Li, “R²-Gaussian: Rectifying radiative Gaussian splatting for tomographic reconstruction,” in *Advances in Neural Information Processing Systems*, 2024.
- [8] J. Batson and L. Royer, “Noise2Self: Blind denoising by self-supervision,” in *Proceedings of the 36th International Conference on Machine Learning*, 2019, pp. 524–533.
- [9] D. A. Koff and H. Shulman, “An overview of digital compression of medical images: can we use lossy image compression in radiology?” *Canadian Association of Radiologists Journal*, vol. 57, no. 4, pp. 211–217, 2006.
- [10] F. Amat, W. Lemon, D. P. Mossing, K. McDole, Y. Wan, K. Branson, E. W. Myers, and P. J. Keller, “Fast, accurate reconstruction of cell lineages from large-scale fluorescence microscopy data,” *Nature Methods*, vol. 11, no. 9, pp. 951–958, 2014.
- [11] M. Dannecker, S. Jia, N. Stolt-Ansó, N. Girard, G. Auzias, F. Rousseau, and D. Rueckert, “Fast and explicit: Slice-to-volume reconstruction via 3D Gaussian primitives with analytic point spread function modeling,” in *Medical Imaging with Deep Learning (MIDL)*, 2026, tODO verify author first names.
- [12] D. S. Taubman and M. W. Marcellin, *JPEG2000: Image Compression Fundamentals, Standards and Practice*. Kluwer Academic Publishers, 2002.
- [13] J. Alakuijala et al., “JPEG XL next-generation image compression architecture and coding tools,” in *Applications of Digital Image Processing XLII*, Proc. SPIE 11137, 2019.
- [14] P. Lindstrom, “Fixed-rate compressed floating-point arrays,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 20, no. 12, pp. 2674–2683, 2014.
- [15] V. Ulman et al., “An objective comparison of cell-tracking algorithms,” *Nature Methods*, vol. 14, pp. 1141–1152, 2017.
- [16] M. Maška et al., “The cell tracking challenge: 10 years of objective benchmarking,” *Nature Methods*, vol. 20, pp. 1010–1020, 2023.
- [17] T. Lindeberg, “Feature detection with automatic scale selection,” *International Journal of Computer Vision*, vol. 30, no. 2, pp. 79–116, 1998.
- [18] C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu, “Cellpose: a generalist algorithm for cellular segmentation,” *Nature Methods*, vol. 18, pp. 100–106, 2021.